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Making and Verifying a Forensic Copy



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UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

Menginovasi Penyelesaian Lestari

DFBOOK DIGITAL FORENSICS · LESSON 3.0

Making and Verifying a Forensic Copy

Lesson overview: copy without changing, then prove it



LESSON 3.0 · OVERVIEW

Making a Forensic Copy

The core skill: copy without changing, then prove the copy is exact

BY THE END YOU WILL BE ABLE TO

- **Explain why an image is not a backup**
  And compare the formats used to store it
- **Plan an acquisition**
  Allowing for SSDs, encryption and failure
- **Image a drive through a write-blocker**
  Under guidance, with FTK Imager
- **Verify a copy with hash values**
  And decide what to do if they differ
- **Explain why tools are validated**
  And how validation differs from verification

LESSON 3.0 · ROUTE

Five Sub-Lessons

Two demos, one desk exercise and two hands-on labs

3.1	**Bit-Stream Images and Storage Formats**	Demo: one image, two formats
3.2	**Planning an Acquisition**	Desk: four scenarios
3.3	**The Workstation, Write-Blockers and Imaging**	Lab: image a training USB
3.4	**Verifying the Copy with Hash Values**	Lab: hash, change, hash again
3.5	**What Forensic Tools Do, and Why We Validate Them**	Demo: two tools, one test image

Labs use instructor-issued training USB drives only. Never connect your own devices or real evidence.

LESSON 3.0 · THE CASE

The Teratai Case So Far

From the evidence locker to the forensic lab

SO FAR

The laptop, USB drive and phone were found, bagged, tagged and logged at the accounts office.

NOW

The lab must copy the drives without changing them, and prove each copy is exact.

IN THE EVIDENCE LOCKER

- **The executive's laptop**
- **The USB drive from the desk**
- **The mobile phone (a later lesson)**
- **Custody forms for each item**

The question for this lesson: how do you examine evidence without changing it, and prove you did not?

UP NEXT

Bit-Stream Images and Storage Formats

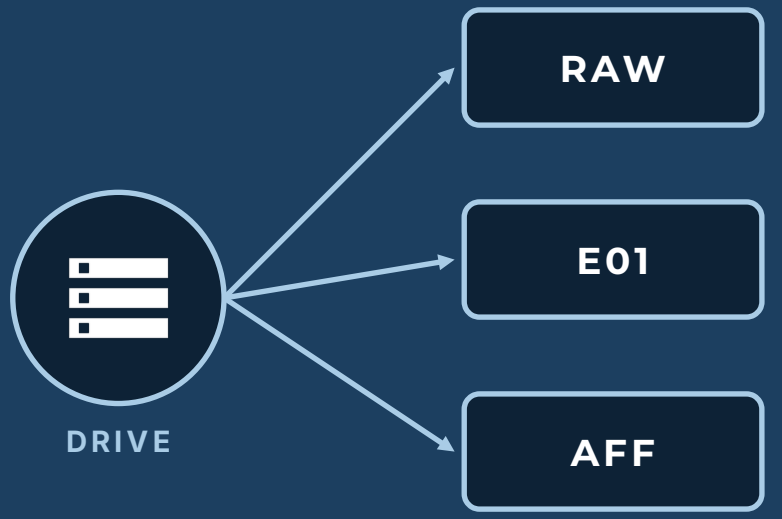
We start with what a forensic copy is, and the file formats that hold it.

- **An image, not a backup**
- **Raw, proprietary and AFF formats**
- **Choosing a format**

DFBOOK DIGITAL FORENSICS · LESSON 3.1

Bit-Stream Images and Storage Formats

What a forensic copy is, and the files that hold it



IN THIS LESSON

1. Image, not backup
2. Storage formats
3. Choosing a format

LESSON 3.1 · OVERVIEW

Bit-Stream Images and Formats

Examiners never work on the original evidence. They make an exact copy, called an image, and examine that. This lesson explains what an image is, and the formats that store it.

BY THE END YOU WILL BE ABLE TO

1. **Explain why a forensic image is not a backup**
2. **Describe raw, proprietary and AFF formats**
3. **Choose and justify a format for a case**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Illustrative: two copies of the same Teratai USB drive

The USB drive

3 files, plus 2 deleted invoices

A backup copy

Copies the 3 current files. The deleted invoices are not there.

✕ MISSES DATA

A bit-stream image

Copies every bit: the 3 files, the deleted invoices and any fragments.

✓ COPIES ALL

Deleted data stays on a drive until it is overwritten. Only an image keeps it for the examiner.

INTRODUCTION · ROADMAP

Your Route Through Lesson 3.1

Each part answers one question, in order

1

PART 1

Image, not backup

How does a forensic copy differ from a backup?

2

PART 2

Storage formats

Which file formats can hold a forensic image?

3

PART 3

Choosing a format

Which format suits a given case?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Image, Not Backup

What a bit-stream copy is, and what a backup leaves out

01

IN THIS PART YOU WILL LEARN TO

- Define a bit-stream copy and an image
- Explain what a backup misses

KEY TERMS

Bit-stream copy **Image** **Target drive**

PART 1 · COMPARISON

Backups and Bit-Stream Copies

The same drive, copied two ways

	Backup copy	Bit-stream copy
What it copies	Files in folders, or of known types	**Every bit, sector by sector**
Deleted files and emails	Not copied	**Copied**
File fragments	Not recovered	**Copied, ready to examine**
Used in forensics?	No	**Yes: the evidence copy**

Bit-stream copy: a bit-by-bit copy of a drive or other storage medium, also called a forensic copy. It is an exact duplicate.

PART 1 · SEQUENCE

From Original to Image

How a bit-stream image is made



Original drive

The evidence. It is only read, never written to.

Write-blocker

Lets data out, and stops anything going in.

Imaging tool

Reads every sector and writes it to a file.

Image file

Saved on a clean target drive.

Image: the file that holds the bit-stream copy of a drive or partition. Also called an image file, mirror or sector copy.

Target drive: the drive the image is saved to. Some older tools need one identical to the original; most current tools do not.

PART 1 · RULES

The First Rule of Digital Forensics

Three habits that protect every case

Preserve the original

Seal it, log it and store it. It is not examined directly.

Work only on the copy

Every search and recovery runs on the image.

Block writes when copying

Acquisition tools need a write-blocker for drives and USB devices.

If the copy is damaged, make another from the original. If the original is damaged, nothing can replace it.

PART 1 · WORKED EXAMPLE

Copying the Executive's Laptop

Illustrative: an IT officer offers a backup they made last week

THE OFFER

"We back up every laptop weekly. Can you use last week's backup instead of copying the drive?"

QUESTION

Is the backup enough?

1. **What does a backup hold?**
   Only current files of known types.
2. **What might matter here?**
   Invoices deleted before the audit.
3. **When was it made?**
   A week ago: newer changes are missing.

ANSWER

No. Keep the backup as extra evidence, but make a bit-stream image of the laptop drive itself.

PART 2 OF 3

Storage Formats

The three families of image file, and what each offers

02

IN THIS PART YOU WILL LEARN TO

- Describe the raw format
- Describe proprietary formats and AFF

KEY TERMS

Raw **E01** **AFF** **Segment**

PART 2 · CONCEPT

The Raw Format

The oldest, simplest image: one flat copy of every bit

ADVANTAGES

Fast and universal

- Fast data transfer
- Can ignore minor read errors on the source
- Almost every forensic tool can read it

DISADVANTAGES

Large and bare

- As large as the original: no compression
- Some free tools may skip weak sectors
- No room inside for case notes or hashes

Raw format: a simple, sequential file of a drive's data, made by tools such as the Linux dd command or FTK Imager.

PART 2 · CATEGORIES

Proprietary Formats

Vendors' own formats add three useful features

Compression

Smaller image files, saving space on the target drive.

Segments

The image split into smaller files, each with its own integrity check.

Metadata inside

Date and time, the original's hash, examiner name and case notes.

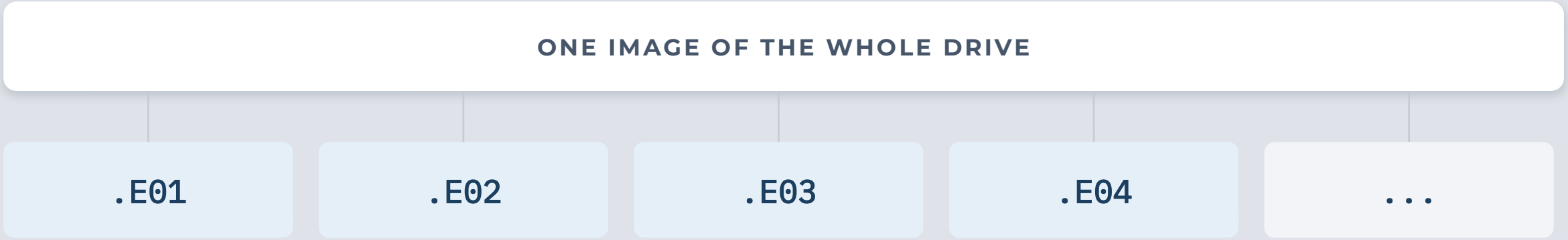
⚠ The drawback: an image in one vendor's format may not open in another vendor's tool.

E01: the Expert Witness format, EnCase's default and the unofficial standard. Its files run .E01, .E02 and so on.

PART 2 · STRUCTURE

Segmented Image Files

One large image, saved as a numbered series of smaller files



Typical size	650 MB per segment, adjustable
Usual maximum	2 GB, because FAT target drives cannot hold larger files
Raw images too	Can be split with Linux tools; rename the parts .01, .02 for other tools

PART 2 · CONCEPT

Advanced Forensic Format (AFF)

An open-source format, designed by Dr Simson Garfinkel, with six goals

Compressed or not
The examiner chooses

No size limit
For any disk-to-image file

Room for metadata
In the image or its segments

Simple and extensible
Easy to build on

Open, any platform
Free for any vendor to use

Self-checking
Built-in consistency checks

File extensions: .afd for segmented image files, and .afm for AFF metadata.

Because AFF is open source, no vendor is restricted from supporting it.

PART 3 OF 3

Choosing a Format

Matching the format to the needs of the case

03

IN THIS PART YOU WILL LEARN TO

- Compare the three formats side by side
- Choose and justify a format

KEY TERMS

Compression **Metadata** **Validation file**

PART 3 · COMPARISON

The Three Formats Compared

The same four questions, asked of each format

	Raw	Proprietary (E01)	AFF
Compression	No: an extra step	Yes, optional	Yes, optional
Hash and notes inside	No: keep a separate file	Yes	Yes
Opens in other tools	Almost all	Many read E01	Any vendor may add it
Ownership	Open	Vendor's own	Open source

Validation file: a separate text file holding a raw image's hash value. Without it, a raw image cannot be checked later.

PART 3 · WORKED EXAMPLE

A Format for the Laptop Image

Illustrative: three needs for the Teratai laptop image

NEED 1

Share it

A second examiner, using a different tool, will check the work.

NEED 2

Keep details with it

The hash and case details should travel inside the image.

NEED 3

Save space

The target drive is barely larger than the laptop drive.

Reasoning: raw is widely read but large and holds no details. E01 is compressed, carries its hash and details, and many tools open it.

ANSWER

E01 meets all three needs. Confirm first that the second examiner's tool can open it.

LESSON 3.1 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Image, not backup

A bit-stream image copies every bit, including deleted files and fragments. A backup does not.

2

Storage formats

Raw is fast and universal; E01 adds compression, segments and metadata; AFF is open source.

3

Choosing a format

Match the format to the case: sharing, details carried inside, and space.

Preserve the original; work only on the copy.

UP NEXT

Planning an Acquisition

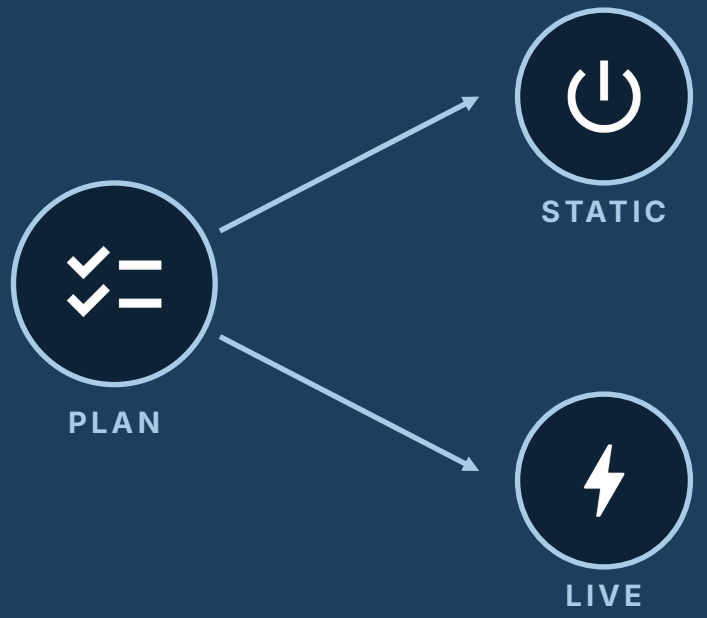
Next: how to prepare for a copy, choose the method, and plan for what could go wrong.

- **Preparing to acquire**
- **Choosing a method**
- **Planning for problems**

DFBOOK DIGITAL FORENSICS · LESSON 3.2

Planning an Acquisition

Prepare, choose the method, and plan for problems



IN THIS LESSON

1. Preparing to acquire
2. Choosing a method
3. Planning for problems

LESSON 3.2 · OVERVIEW

Planning an Acquisition

You may get only one chance to copy a drive, sometimes against the clock. Planning picks the right method, and prepares for what could go wrong.

BY THE END YOU WILL BE ABLE TO

1. **Plan an acquisition before you start**
2. **Choose and justify an acquisition method**
3. **Allow for SSDs, encryption and failure**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Case study: the laptop seized while it was open (United States, 2013)



Encrypted laptop

Closed, its contents were unreadable

A plan around it

Agents waited until he was logged in

Seized while open

Taken before it could be locked

When federal agents arrested the operator of the Silk Road website in a public library, they took his laptop while it was still unlocked.

The method was chosen before anyone reached the scene. That is acquisition planning.

INTRODUCTION · ROADMAP

Your Route Through Lesson 3.2

Each part answers one question, in order

1

PART 1

Preparing to acquire

What must you know before you start?

2

PART 2

Choosing a method

Which way of copying fits the case?

3

PART 3

Planning for problems

What could go wrong, and how do you prepare?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Preparing to Acquire

What to find out, and what to write down, before you copy

01

IN THIS PART YOU WILL LEARN TO

- Ask the seven planning questions
- Write an acquisition action plan

KEY TERMS

Static acquisition **Live acquisition**

PART 1 · METHOD

Seven Questions Before You Copy

The more you know about the drive, the fewer surprises

1. **When and where will it be done?**
2. **Is the drive loose, or inside a computer?**
3. **Can it be removed easily?**
4. **Can you keep it, or must it go back?**
5. **What size and make is it?**
6. **If it stays in, can you reach the boot menu?**
7. **Do you have the resources?**

Static acquisition

A copy made from a drive that is switched off. Always the preferred way.

Live acquisition

A copy made from a running system, for example when its data is readable only while logged in.

PART 1 · CATEGORIES

The Acquisition Action Plan

Four elements in every plan; the details change with each case

Resources

Connectors, cables and target media

Outline

How the copy will be made, step by step

Preservation

How the original stays unchanged

Security

Before, during and after the copy

Record serial numbers

Of the source and target drives, before copying starts.

Keep continuous control

Away from the lab, an authorised person stays with the copy at all times.

PART 2 OF 3

Choosing a Method

Four ways to copy, and how much to copy

02

IN THIS PART YOU WILL LEARN TO

- Compare the four acquisition methods
- Choose a full, logical or sparse copy

KEY TERMS

Disk-to-image **Logical** **Sparse**

PART 2 · COMPARISON

Four Acquisition Methods

Each works for both static and live acquisitions

Method	What it makes	When it is used
Disk-to-image file	Image files of the whole drive	**Most common and most flexible**
Disk-to-disk copy	A copy on a second drive	When an image cannot be made, often on older drives
Partition-to-partition	A copy of one partition	When only one partition is needed
Partition-to-data file	A copy of files and folders	A simple copy of chosen data

Disk-to-image file: one or many bit-for-bit copies of a drive, which most tools can read as if it were the drive.

PART 2 · CONCEPT

How Much to Copy

From everything on the drive to only the files you need

MORE DATA, MORE TIME → LESS DATA, LESS TIME

Physical (full)

Every bit on the drive

USE
The default whenever time and space allow

Sparse

Chosen files, plus fragments of deleted data

USE
Only when the whole drive is not needed

Logical

Only chosen files or file types

USE
Email files only; a huge server or storage network

Logical acquisition: copies only specific files of interest. Increasingly preferred for very large systems, such as in civil disclosure (e-discovery).

PART 2 · METHOD

Choosing the Method

Four factors decide which method fits

Size of the source

A large drive needs an equally large target, or compression.

Keep or return

If the drive must go back to its owner, the copy must be good.

Time available

Large drives can take hours; logical or sparse copies are faster.

Location

In the lab, or at a remote site with less equipment and security.

One chance: in civil disclosure cases, you often may not keep the original. Check the copy before the drive is returned.

PART 2 · CONTRAST

Compressing an Image

Saving space without changing a single bit

LOSSLESS

Used in forensics

Every bit is restored exactly. Can save up to half the space, but zip files hardly shrink.

LOSSY

Never used in forensics

Throws data away to save space, as in JPEG photos. It changes the original data.

TEST IT: THE SAME HASH BEFORE AND AFTER

1 **Hash the file** → 2 **Compress it** → 3 **Uncompress it** → 4 **Hash it again**

If the hashes differ, a hardware or software error has corrupted the compressed file.

PART 2 · WORKED EXAMPLE

How Long Will the Copy Take?

Illustrative: a 500 GB drive copied over a USB 2.0 link

THE SET-UP

Source drive

500 GB, reads at 6 Gbps

Link (USB 2.0)

480 Mbps: the slowest

Target drive

Writes at 4 Gbps

1. **Size in bits**

   `500 GB x 8 = 4,000,000 megabits`

2. **Divide by the slowest rate**

   `4,000,000 / 480 = about 8,300 seconds`

3. **Convert to hours**

   `8,300 / 3,600 = about 2.3 hours`

ANSWER

At least 2.3 hours. Allow extra for verifying, and for slow processors or poor cables.

PART 3 OF 3

Planning for Problems

Failure, hidden areas, encryption and SSDs

03

IN THIS PART YOU WILL LEARN TO

- Plan for failure and hidden areas
- Allow for encryption and SSDs

KEY TERMS

HPA | Whole disk encryption | TRIM

PART 3 · METHOD

Contingency Planning

If anything can go wrong, it will: four precautions

Make at least two images	With two different tools if you can; tools differ in how hard they retry damaged areas.
Check for the hidden area	The host protected area (HPA) is hidden from the operating system; some tools miss it.
Carry spare, new cables	Faulty cables and connectors are a common cause of failed copies.
Expect media to fail	Every drive has a mean time to failure (MTTF), set by its maker.

PART 3 · CONCEPT

Encrypted Drives

Whole disk encryption locks every sector of the drive

COPIED WHILE LOCKED

Encrypted data only

- A static copy holds only unreadable data
- The examiner still needs the key to read it

WITH THE KEY

Decrypt, then copy

- Decrypting can take several hours
- It does not change allocated or deleted data

Whole disk encryption: encrypts every sector of a drive, as BitLocker does in Windows. A suspect may refuse the key.

In Malaysia, Criminal Procedure Code section 116B covers access to computerised data in a search, including passwords and decryption. Confirm with a legal adviser.

PART 3 · STRUCTURE

Solid-State Drives (SSDs)

Flash memory that tidies itself, even after deletion

Cells → Pages → **Blocks** → Planes → Dies → Chip

Data is written a page at a time, but a whole block must be erased before it can be written again.

Wear levelling

Spreads writes evenly so that every cell wears at the same rate.

Garbage collection

Moves data and erases old blocks in the background.

TRIM

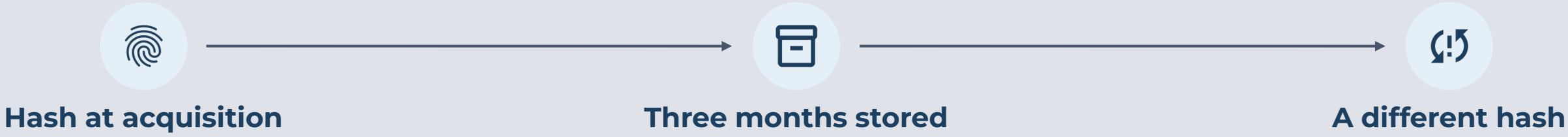
The operating system tells the drive which pages are free to erase.

⚠ **Deleted data may be erased by the drive itself. Acquire an SSD as soon as possible.**

PART 3 · WORKED EXAMPLE

The SSD Whose Hash Changed

Illustrative: an SSD hashed again after three months in the locker



1 Why did it change?

Garbage collection and TRIM erased unused areas while it was stored.

2 What should the lab do?

Consider a policy to disable TRIM before storage, with legal advice, and record it.

ANSWER

The image verified at acquisition is still the evidence copy. Explain the change in your notes.

LESSON 3.2 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Preparing to acquire

Seven questions, then a plan: resources, outline, preservation and security.

2

Choosing a method

Disk-to-image is most common. Weigh size, keep or return, time and location.

3

Planning for problems

Two images, the hidden area, spare cables, encryption keys, and SSDs acquired early.

Plan the copy before you touch the drive.

UP NEXT

The Workstation, Write-Blockers and Imaging

Next, a hands-on lab: image a training USB drive through a write-blocker.

- **The forensic workstation**
- **Preparing the target drive**
- **Imaging with FTK Imager**

DFBOOK DIGITAL FORENSICS · LESSON 3.3

The Workstation, Write-Blockers and Imaging

Your first forensic image, made in the lab

IN THIS LESSON

1. The forensic workstation
2. Preparing the target drive
3. Imaging with FTK Imager



LESSON 3.3 · OVERVIEW

Write-Blockers and Imaging

In this lab you make your first forensic image: a training USB drive, copied through a hardware write-blocker with FTK Imager.

BY THE END YOU WILL BE ABLE TO

1. **Explain what a write-blocker does**
2. **Prepare and record a target drive**
3. **Image a drive with FTK Imager, under guidance**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Connecting a drive is never neutral

Plug it in

The drive is connected to an ordinary computer.

The system takes over

It can write to any drive it can reach.

Small changes

System files added, dates updated.

In court: "Can you prove nothing on this drive changed after you received it?"

A write-blocker lets you answer yes: data can be read, but nothing is written.

INTRODUCTION · ROADMAP

Your Route Through Lesson 3.3

Each part answers one question, in order

1

PART 1

The forensic workstation

What protects the evidence while it is copied?

2

PART 2

Preparing the target drive

How is the destination made ready?

3

PART 3

Imaging with FTK Imager

How is the image made in the lab?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

The Forensic Workstation

Write-blockers, and the computer they connect to

01

IN THIS PART YOU WILL LEARN TO

- Compare software and hardware write-blockers
- List what a forensic workstation needs

KEY TERMS

Write-blocker **Hot-swappable**

PART 1 · COMPARISON

Two Kinds of Write-Blocker

Both stop writes to the evidence drive, in different ways

SOFTWARE

Settings in the system

- Sets chosen drives to read-only
- WinFE asks which drives to protect at start-up
- Linux forensic modes leave drives unmounted

HARDWARE

A device in between

- A bridge between the drive and the workstation
- Start the system as usual
- Ideal for graphical forensic tools

Write-blocker: a device or program that prevents data being written to an evidence drive. Check any built-in software blocker before trusting it.

PART 1 · SEQUENCE

What a Hardware Write-Blocker Does

A write that seems to work, but never reaches the drive

A program saves a file

Windows reports that the copy succeeded.

→

The blocker steps in

It discards the data: it is written to null.

→

The drive is unchanged

After a restart, the file is not there.

Hot-swappable: most write-blockers let you disconnect one drive and connect the next without shutting down the workstation.

They connect through USB, SATA and other interfaces. Test results for write-blockers are published by NIST.

PART 1 · WORKED EXAMPLE

Is the Write-Blocker Working?

Illustrative: a check on a test drive before the Teratai images are made

THE TEST

A test drive, with no evidence on it, is connected through the lab's write-blocker.

QUESTION

How do you show it blocks writes?

1. **Copy a file to the test drive**
   Windows reports success.
2. **Restart, and reconnect the drive**
   Through the same write-blocker.
3. **Look for the file**
   It is not there: the write was discarded.

ANSWER

It works. Record the test, so you can show the blocker was checked before use.

PART 1 · CATEGORIES

The Forensic Workstation

What a lab computer for forensic work should include

Full tower case

Room to add drives and cards

Memory and processor

As much as the budget allows

Backed-up power

A strong power supply, with a battery

Spare cables

Extra power and data cables

Drive adapters

To connect older and newer drives

External ports

USB and other connections

Good display

Two monitors for heavy use

In the field

A laptop and a portable write-blocker

Whatever you choose, make sure it does what your cases need.

PART 2 OF 3

Preparing the Target Drive

A clean, recorded drive to hold the image

02

IN THIS PART YOU WILL LEARN TO

- Prepare a clean target drive
- Record every drive used in the copy

KEY TERMS

Wiping **Commingling**

PART 2 · METHOD

Preparing a Target Drive

Three steps, and the reason behind them

1. **Use a new or wiped drive**

   Freshly wiped and formatted, even if it is new.

2. **Record its details**

   Serial number, maker, size in bytes, partition labels.

3. **Secure it**

   Keep it ready, and safe, until it is used.



WHY IT MATTERS

The other side's lawyers may have their own examiner search your target drive. Old data from another case could be used to cast doubt on your work.

Commingling: evidence data mixed with data from elsewhere, such as an earlier case.

PART 2 · METHOD

Wiping and Formatting

Two Windows commands, and a choice of file system

format with /p:0

```
format k: /fs:fat32 /p:0
```

Writes zeros over one partition, in one pass. Recent Windows allows it only on USB flash drives.

diskpart clean all

```
select disk 2   then   clean all
```

Wipes a whole drive and its partitions. Then create a partition and format it. Large drives take hours.

⚠ **Check the drive before you wipe it. Wiping the wrong drive, such as C:, cannot be undone.**

File system: FAT32 or exFAT for small targets; NTFS for large ones.

PART 2 · WORKED EXAMPLE

Which Disk Is the Target?

Illustrative: the list of disks a wiping tool shows

Disk	Size	Status
Disk 0	931 GB	Online
Disk 1	232 GB	Online
Disk 2	7.5 GB	Online

Your record for the target: an 8 GB USB drive, labelled TARGET-02.

1. **Which disk holds Windows?**
   Disk 0: the largest, the C: drive.

2. **Which size matches?**
   Disk 2: 8 GB drives show as about 7.5 GB.

3. **How do you confirm?**
   Check the serial number in Disk Management.



ANSWER

Disk 2, but only after its serial number matches your record.

PART 2 · STRUCTURE

Recording the Media

Before any copy, record four details for three kinds of media

	Maker or label	Model	Serial number	Size
Boot media	✓	✓	✓	✓
Source drive	✓	✓	✓	✓
Target drive	✓	✓	✓	✓

Boot media: a CD, DVD or flash drive used to start a computer for an acquisition, such as a forensic boot disk.

USB flash drives often have no printed label. A Windows system tool can show the details instead.

PART 3 OF 3

Imaging with FTK Imager

The tool, and the steps you will follow in the lab

03

IN THIS PART YOU WILL LEARN TO

- Describe what FTK Imager does
- Follow the lab steps to image a drive

KEY TERMS

Physical image **Verify**

PART 3 · CONCEPT

What FTK Imager Does

A small, free acquisition program that runs in Windows

Copies drives

Physical or logical bit-by-bit images, or chosen files and folders (sparse).

Hashes as it copies

Calculates MD5 and SHA-1 of the source, then checks the image against them.

Keeps the record

Hashes and notes go into a separate file, and inside an E01 image.

⚠ **FTK Imager does not block writes itself. Use a hardware write-blocker, or a forensic boot disk such as Mini-WinFE.**

PART 3 · LAB

Imaging the Training USB

Case thread: the training copy of the Teratai USB drive

1 **Record**

Note the details of the source and target drives.

2 **Connect**

Training USB to the write-blocker, then to the workstation.

3 **Create the image**

In FTK Imager: Create Disk Image, then Physical Drive.

4 **Describe it**

Choose E01, and enter the case, evidence number and your name.

5 **Set the target**

Pick the target folder, and tick the option to verify the image.

6 **Start and check**

When it ends, check the verification result and save the report.

⚠ **Use only the training USB and target drive your instructor issues. Never connect your own devices.**

LESSON 3.3 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

The forensic workstation

Software or hardware write-blockers stop writes; test the blocker before use.

2

Preparing the target drive

New or wiped, recorded and secured. Check twice before wiping anything.

3

Imaging with FTK Imager

Through a write-blocker, as an E01, with verification turned on.

Copy from the evidence; write only to the target.

UP NEXT

Verifying the Copy with Hash Values

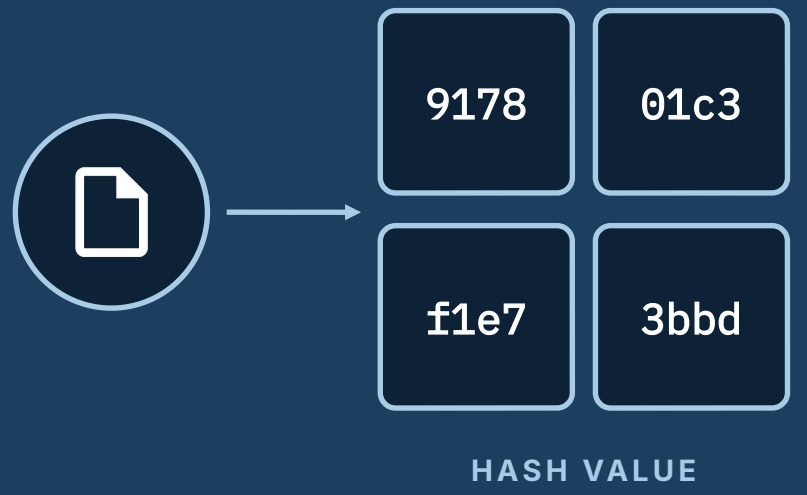
Next, a second lab: prove your image is exact, and see what one changed byte does.

- **What a hash value is**
- **Computing hashes**
- **When hashes do not match**

DFBOOK DIGITAL FORENSICS · LESSON 3.4

Verifying the Copy with Hash Values

Proving that a copy is exact



IN THIS LESSON

1. What a hash value is
2. Computing hashes
3. When hashes do not match

LESSON 3.4 · OVERVIEW

Verifying the Copy

A copy is only useful in court if you can prove it is exact. Hash values give that proof, and they show at once when something has changed.

BY THE END YOU WILL BE ABLE TO

1. **Explain what a hash value is**
2. **Compute and compare hash values**
3. **Decide what to do when hashes do not match**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Two texts that differ by one capital letter, and their MD5 hash values

TEXT		MD5 HASH VALUE
`Teratai Supplies`	→	`917801c3f1e73bbd0feb4a059822ca75`
`teratai Supplies`	→	`4aa0d9e3aabdd353d0a3568b220aeb1f`

One small change gives a completely different value. So if the hash of a copy matches the original, the copy has not changed.

A matching hash is mathematical proof that the evidence did not change.

INTRODUCTION · ROADMAP

Your Route Through Lesson 3.4

Each part answers one question, in order

1

PART 1

What a hash value is

How can one number prove a copy is exact?

2

PART 2

Computing hashes

Which tools give you a hash value?

3

PART 3

When hashes do not match

What do you do when the values differ?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

What a Hash Value Is

A digital fingerprint for a file or a whole drive

01

IN THIS PART YOU WILL LEARN TO

- Explain what a hash value shows
- Compare the common hash algorithms

KEY TERMS

Hash value **Algorithm** **Collision**

PART 1 · CONCEPT

A Digital Fingerprint

Any amount of data in, one short fixed-length value out

Data
A file, a folder or a whole drive

→

Algorithm
A fixed mathematical formula

→

Hash value
A string of hexadecimal characters

THREE RULES FOR A FORENSIC HASH

1. You cannot predict the hash value of a file or drive.
2. No two different files should share a value. Rare exceptions are called collisions.
3. If anything in the file or drive changes, the value changes.

Hash value: a unique value calculated from data, used to show it has not changed.

PART 1 · COMPARISON

Common Hash Algorithms

From a simple check to the stronger modern algorithms

Algorithm	What to know
CRC-32	Detects changes, but is not a forensic hash
MD5	The first forensic hash; still widely used, though collisions exist
SHA-1	Developed by NIST; used for the national library of known files
SHA-256, SHA-512	Newer and stronger; increasingly the standard



Good practice: record two algorithms for every acquisition, such as MD5 and SHA-256.

PART 1 · SEQUENCE

Verifying a Copy

Four steps that turn a copy into trusted evidence



Hash the original

During acquisition, through the write-blocker.

Hash the image

Most imaging tools do this for you.

Compare

Every character of the two values.

Record

Both values, in your notes and on the custody form.

Then check again: hash the image before each analysis, and compare with the value recorded at acquisition.

Nonkeyed hash: a hash any tool can calculate. It identifies a known file even if the file is renamed.

PART 2 OF 3

Computing Hashes

Commands and tools that calculate a hash value

02

IN THIS PART YOU WILL LEARN TO

- Hash a file or drive on Linux and Windows
- Know where an image's hash is kept

KEY TERMS

md5sum **Get-FileHash** **Validation file**

PART 2 · METHOD

Hashing on Linux and Windows

Both systems have built-in commands

Linux

Hash one file

```
md5sum invoice.pdf
```

Hash a whole drive

```
sha256sum /dev/sdb
```

Save the result

```
md5sum invoice.pdf > hash.txt
```

Windows PowerShell

Hash one file

```
Get-FileHash invoice.pdf
```

Choose an algorithm

```
... -Algorithm SHA256
```

Save the result

```
Get-FileHash t*.* > hash.txt
```

⚠ Free stand-alone hashing tools exist, but download them with care: some carry malware or adware.

PART 2 · CONTRAST

Where the Hash Is Kept

Raw and E01 images store their hash values differently

RAW IMAGE

In a separate file

- The image has no room for metadata
- Keep the validation file the tool creates
- Use it later to check the image is still good

E01 IMAGE

Inside the image

- The MD5 is written into the first segment
- FTK Imager also saves a report with MD5 and SHA-1
- Many tools re-check it and report a mismatch

Validation file: a text file holding the hash of an acquisition. For a raw image, it is the only record.

PART 2 · WORKED EXAMPLE

Do These Values Match?

Illustrative: two MD5 values, from an original and its image

ORIGINAL `c4ca4238a0b923820dcc509a6f75849b`

IMAGE `c4ca4238a0b923820dcc509a6f75a94b`

1 **Check the start?**

The first 28 characters are the same.

2 **Check the whole value**

Characters 29 and 30 differ: 84 and a9.

3 **Let a tool compare**

Paste both into a tool, rather than reading by eye.

ANSWER

They do not match. A value that is almost the same is still a mismatch.

PART 2 · LAB

Hash, Change, Hash Again

Lab: verify your image, then see what one changed byte does

1. **Hash the source**
   The training USB, through the write-blocker.
2. **Hash the image**
   Your E01 from Lesson 3.3; compare all three values.
3. **Make a scratch copy**
   Export one file to a scratch folder, and hash it.
4. **Change one byte**
   Open the scratch copy in a hex editor, such as HxD.
5. **Hash it again**
   Compare the two values for the scratch copy.
6. **Record it all**
   Every value, the tool used, and what you saw.

⚠ **Change only the scratch copy. Never edit the image, the training USB or any evidence.**

PART 3 OF 3

When Hashes Differ

Finding the cause, and deciding what to do

03

IN THIS PART YOU WILL LEARN TO

- Find the likely cause of a mismatch
- Decide and record what to do next

KEY TERMS

Mismatch **Byte-by-byte** **MTTF**

PART 3 · CATEGORIES

Why Hashes Might Not Match

Most mismatches have an ordinary cause; some do not

A faulty cable

Connectors fail most, because they are plugged and unplugged.

Failing media

Drives wear out; each has a mean time to failure (MTTF).

SSD housekeeping

Garbage collection or TRIM erased unused areas.

A tool or system error

For example, while compressing the image.

A real change

The data has been altered, by accident or on purpose.

Mismatch: not proof of tampering. It means: find out what differs, and why.

PART 3 · METHOD

What to Do About a Mismatch

Four steps, in order, all written down

1. **Record it** — FIRST
   Note both values, the time and the tools. Delete nothing.
2. **Copy again** — IF POSSIBLE
   If you still have the original, re-image it and compare again.
3. **Suspect the media** — IF IT FAILS AGAIN
   If the new copy also fails to match, the original drive may be failing.
4. **Find the difference** — THEN
   Compare the files byte by byte, and judge how it affects the case.

Byte-by-byte comparison: checks every byte of two files, with commands such as comp in Windows or diff in Linux.

PART 3 · WORKED EXAMPLE

The Laptop Image That Did Not Match

Illustrative: the image's hash differs from the source's

THE PROBLEM

The imaging log shows read errors, and the USB cable was found loose afterwards.

QUESTION

What should the examiner do?

1. **Record the mismatch**
   Both values, the log and the loose cable.
2. **Fit a new cable**
   Old cables are thrown away, not reused.
3. **Re-image from the original**
   This time, the hashes match.

ANSWER

Use the second, verified image, and report both attempts openly.

LESSON 3.4 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

What a hash value is

A digital fingerprint: any change to the data gives a different value.

2

Computing hashes

md5sum and sha256sum on Linux; Get-FileHash on Windows; E01 stores its own.

3

When hashes do not match

Record it, copy again, suspect the media, then find what differs.

No matching hash, no trusted copy.

UP NEXT

What Forensic Tools Do, and Why We Validate Them

Next: the jobs forensic tools do, and how labs prove the tools work.

- **What tools do**
- **Validation and verification**
- **Testing your tools**

DFBOOK DIGITAL FORENSICS · LESSON 3.5

What Forensic Tools Do, and Why We Validate Them

Knowing your tools, and proving they work



IN THIS LESSON

1. What tools do
2. Validation and verification
3. Testing your tools

LESSON 3.5 · OVERVIEW

Forensic Tools and Validation

Every finding depends on the tool that produced it. Examiners must know what their tools do, and be able to prove that the tools work.

BY THE END YOU WILL BE ABLE TO

1. Name the five functions of forensic tools
2. Explain validation and verification
3. Describe how a lab tests its tools

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Case study: one web visit, reported as 84 (United States, 2011)

84 visits, said one tool's report

1 visit, showed a second tool

At a murder trial in Florida, a report said a web page about chloroform had been visited 84 times. The first tool's developer later found his report was wrong.

A tool's output is not automatically true: check it with a second tool.

INTRODUCTION · ROADMAP

Your Route Through Lesson 3.5

Each part answers one question, in order

1

PART 1

What tools do

What jobs does a forensic tool do?

2

PART 2

Validation and verification

How do the two checks differ?

3

PART 3

Testing your tools

How does a lab prove a tool works?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

What Forensic Tools Do

Five functions shared by every forensic tool

01

IN THIS PART YOU WILL LEARN TO

- Name the five tool functions
- Describe acquisition in more detail

KEY TERMS

Extraction **Reconstruction**

PART 1 · CATEGORIES

The Five Functions

Used to group and evaluate forensic tools, hardware and software

1



Acquisition

Copying the original drive or data.

2



Validation and verification

Proving tools and copies can be trusted.

3



Extraction

Recovering and finding data in the copy.

4



Reconstruction

Rebuilding a drive or files as they were.

5



Reporting

Recording what was found, and how.

Standards say first responders should use validated tools, and record what was done, and why.

PART 1 · STRUCTURE

Acquisition in Detail

The first function, broken into its parts

- ✓ **Physical data copy**
- ✓ **Logical data copy**
- ✓ **Data acquisition format**
- ✓ **Command-line acquisition**
- ✓ **Graphical (GUI) acquisition**
- ✓ **Remote, live and memory acquisition**

Physical or logical: a physical copy takes the whole drive; a logical copy takes one partition. A logical copy of a running, logged-in system may be the only way to read an encrypted drive.

Hardware imagers copy drives on their own and report the bytes copied and the hash, but analysis still needs software.

PART 1 · WORKED EXAMPLE

Five Tasks in the Teratai Case

Illustrative: which function does each task use?

Task	Function
Copy the laptop drive	Acquisition
Prove the image matches the drive	Validation and verification
Recover the deleted invoices	Extraction
Show how the drive looked to a court	Reconstruction
Write up the findings	Reporting



One case uses all five. Each function needs a tool that has been checked.

PART 2 OF 3

Validation and Verification

Two checks that sound alike, but test different things

02

IN THIS PART YOU WILL LEARN TO

- Tell validation from verification
- Explain repeatable and reproducible results

KEY TERMS

Validation **Verification** **Filtering**

PART 2 · CONTRAST

Validation and Verification

One checks the tool; the other checks the data

VALIDATION

Does the tool work?

- Confirms a tool does what it should
- Tested on reference images with a known answer list
- Done before the tool is trusted

VERIFICATION

Is the data identical?

- Proves two sets of data are the same
- Usually by comparing hash values
- Done on every copy, in every case

Filtering: sorting findings to separate ordinary data from suspicious data. It relies on both checks.

PART 2 · CONCEPT

Repeatable and Reproducible

Two tests of accuracy that every forensic tool must pass

Repeatable

The same tool gives the same result every time it is used on the evidence.

Reproducible

Different tools, or different systems, give the same result on the same evidence.

These come from ISO 5725, a standard for the accuracy of measurement in laboratories.

If two tools disagree, find out which one is right, and why, before you use either result.

PART 2 · WORKED EXAMPLE

One File, Two Systems

Illustrative: is this SHA-256 result reproducible?

WINDOWS

```
Get-FileHash invoice.pdf
```

```
SHA256   5e2b...91c0
```

LINUX

```
sha256sum invoice.pdf
```

```
5e2b...91c0
```

1. **Same file?**
   Yes: the same copy of the invoice.
2. **Same algorithm?**
   Yes: SHA-256 on both.
3. **Same value?**
   Yes: every character matches.

ANSWER

Yes. Two systems agree, so the result is reproducible.

PART 3 OF 3

Testing Your Tools

How labs prove, and keep proving, that tools work

03

IN THIS PART YOU WILL LEARN TO

- Describe how a lab tests a tool
- Apply the validation protocols

KEY TERMS

CFTT | CFReDS | NSRL

PART 3 · SEQUENCE

How a Lab Tests a Tool

Six criteria from NIST, with records kept at every step

1 → 2 → 3

Set categories

Group tools by job, such as email tools.

List requirements

What a tool in each category must do.

Write test assertions

Tests that prove or disprove each one.

4 → 5 → 6

Choose test cases

Data with known answers, such as a closed case.

Set the test method

How the tool will be tested.

Report the results

Accurate, clear and objective (ISO 17025).

NIST: the National Institute of Standards and Technology, in the United States.

PART 3 · CATEGORIES

Free Resources for Testing

Three NIST projects that examiners use

CFTT

Computer Forensics Tool Testing

Test methods, and published results for tools and write-blockers.

CFReDS

Computer Forensic Reference Data Sets

Test images with known contents, for testing and training.

NSRL

National Software Reference Library

SHA-1 hashes of known software, to filter out ordinary files.

The NSRL also helps examiners find known bad files, such as malware.

PART 3 · METHOD

Validation Protocols

Habits that answer: "How did you verify your results?"

EXAMINATION PROTOCOL

1. **Examine the evidence with one graphical tool**
2. **Repeat it with a disk editor, such as HxD**
3. **Hash any recovered file in both, and compare**

UPGRADE PROTOCOL

Test every new release, patch and system update before using it on evidence. If it fails, do not use it, and report the problem to the vendor.

Check what a tool actually does, not what it should do, and record it.

PART 3 · WORKED EXAMPLE

A Tool Update Before the Teratai Analysis

Illustrative: the lab's analysis tool has just been updated

THE TEST

The updated tool is run on a reference image whose answer list shows 12 deleted files.

RESULT

The update finds only 11.

1. **Is the update reliable?**
   No: it missed a file the answer list includes.
2. **Can it be used on evidence?**
   Not until the problem is fixed and retested.
3. **What else is needed?**
   Record the test, and report the fault to the vendor.

ANSWER

Keep using the validated version for the Teratai analysis until the update passes.

LESSON 3.5 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

What tools do

Acquisition, validation and verification, extraction, reconstruction and reporting.

2

Validation and verification

Validation checks the tool; verification checks the data. Results must repeat and reproduce.

3

Testing your tools

NIST's six criteria, reference images, and the examination and upgrade protocols.

Trust the tool only after you have tested it.

UP NEXT

Lesson 3 Summary

Next, a short review brings Lessons 3.1 to 3.5 together before Lesson 4.



Images and formats



Planning and imaging



Hashes and validated tools

DFBOOK DIGITAL FORENSICS · LESSON 3 · SUMMARY

Making a Forensic Copy: Lesson Review

Copy without changing, then prove it, in one short review



IN THIS LESSON

1. Images and formats
2. Planning, write-blockers and imaging
3. Hashes and validated tools

LESSON 3 · REVIEW

Lesson 3 at a Glance

One key point from each sub-lesson

3.1 Images and formats	An image copies every bit, deleted data included. Raw, E01 or AFF.
3.2 Planning	Seven questions, a method to fit, and plans for SSDs, encryption and failure.
3.3 Imaging	A tested write-blocker, a clean target drive, and FTK Imager.
3.4 Hash values	Matching hashes prove the copy; a mismatch is recorded and investigated.
3.5 Tools	Five functions; validate the tool, verify the data.

LESSON 3 · REVIEW

From Drive to Trusted Copy

How the five sub-lessons fit together



Copy without changing, then prove the copy is exact.

UP NEXT

Where Data Lives: Disks and File Systems

Lesson 4 explains why deleted and hidden data can still be found.

- **How disks store data**
- **File systems**
- **Deleted and hidden data**

